

Homework

CKY Parsing algorithm assignment:

show the detailed parses of the sentences "Book the flight through Houston" and "I prefer the flight through Houston" using CKY algorithm and L1 grammar given in the slides, including additional lexicon if needed.

Answer

Book the flight through Houston

L1 grammar

$S \rightarrow VP, PP$

$VP \rightarrow V, NP$

$NP \rightarrow Det$

$PP \rightarrow Preposition (NP)$

	[0,1]	[1,2]	[2,3]	[3,4]	[4,5]
0	V ←		VP ←		S
1		Det ←	NP ↓ N		↓
2				Preposition ←	PP ↓ NP
3					
4					

$V \rightarrow \text{Book}$

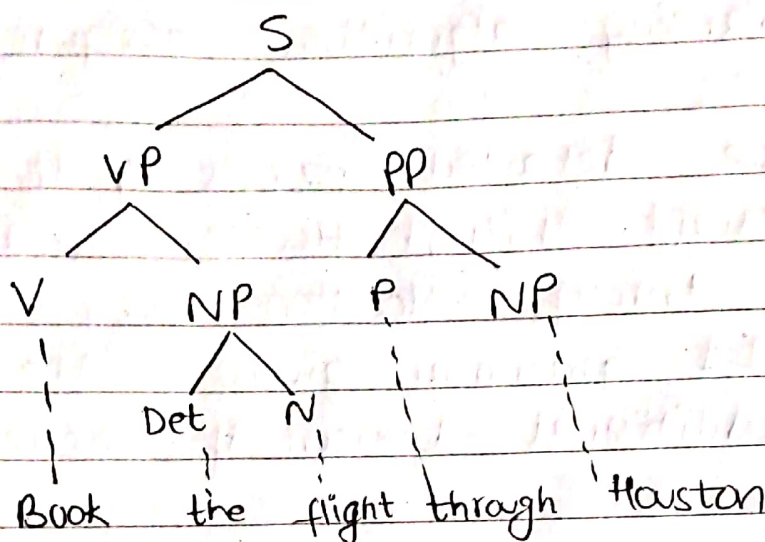
$Det \rightarrow \text{the}$

$N \rightarrow \text{flight}$

$P \rightarrow \text{through}$

$NP \rightarrow \text{Houston}$

Parse tree is :



grammar

$S \rightarrow NP VP$

$VP \rightarrow VNP PP$

$NP \rightarrow Det N$

$PP \rightarrow preposition NP$

Lexicon

$NP \rightarrow I$

$V \rightarrow prefer$

$Det \rightarrow the$

$N \rightarrow flight$

$Preposition \rightarrow through$

$NP \rightarrow Houston$

$S \rightarrow NP VP$

$VP \rightarrow X_1 PP$

$X_1 \rightarrow V NP$

$NN \rightarrow Det N$

$PP \rightarrow P NP$

$NP \rightarrow I$

$V \rightarrow prefer$

$Det \rightarrow the$

$N \rightarrow flight$

$Preposition \rightarrow through$

$NP \rightarrow Houston$

I prefer the flight through Houston

	1	2	3	4	5	6
0	NP ←					S ↓
1		V ←		X ₁ ←		VP ↓
2			Det. ←	NP ↓		
3				N ↓		
4					P ←	PP ↓
5						NP

